AI Chatbot

Desktop AI Chatbot using Groq API

Muhammad Ameer Hamza

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1 Project Introduction

1.1 Background

Artificial Intelligence has transformed the way users interact with systems by enabling intelligent conversational interfaces. Modern chatbots provide instant responses and enhance productivity.

1.2 Problem Statement

Many users require a simple desktop-based AI chatbot without relying on web browsers or complex setups. Existing solutions are either web-based or difficult to integrate for learning purposes.

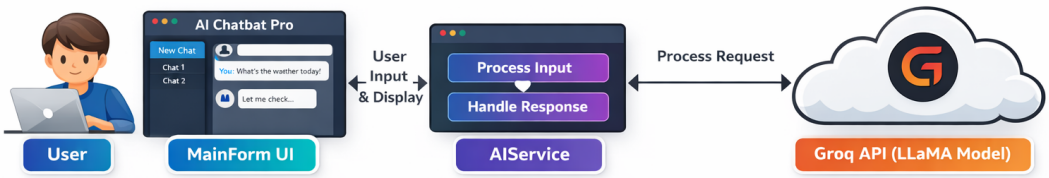
1.3 Objective

The objective of this project is to develop a Windows Forms-based AI chatbot using C# and .NET that:

- Provides a modern and user-friendly interface
- Enables real-time AI interaction
- Uses Groq API for fast AI responses

2 System Design

2.1 Architecture Diagram



2.2 Workflow

1. User enters a message in the input box.
2. Message is displayed in the chat UI.
3. Message is sent to AIService.
4. AIService calls Groq API.
5. Response is received and displayed.

3 Tools and Technologies

- **Framework:** .NET
- **Language:** C#
- **UI:** Windows Forms
- **Libraries:** Newtonsoft.Json
- **API:** Groq API (LLaMA Models)

4 Implementation

4.1 Sending Message and Displaying Response

```
private async Task SendMessage()
{
    string message = txtInput.Text.Trim();
    if (string.IsNullOrEmpty(message)) return;

    AddBubble(message, true);
    txtInput.Clear();

    var typing = AddTyping();

    AIResponse response = await aiService.GetResponse(message);

    chatPanel.Controls.Remove(typing);

    await TypeBubble(response.Content);
}
```

Explanation: This method sends the user's message, shows typing animation, calls the AI API, and displays the response.

4.2 Groq API Integration

```
public async Task<string> GetResponse(string prompt)
{
    var url = "https://api.groq.com/openai/v1/chat/completions";

    var requestBody = new
    {
        model = "llama3-8b-8192",
        messages = new[]
        {
            new { role = "user", content = prompt }
        }
    };

    var json = JsonConvert.SerializeObject(requestBody);

    var request = new HttpRequestMessage(HttpMethod.Post, url);
```

```
    request.Headers.Add("Authorization", $"Bearer {_apiKey}");
    request.Content = new StringContent(json);

    var response = await _client.SendAsync(request);
    var result = await response.Content.ReadAsStringAsync();

    dynamic data = JsonConvert.DeserializeObject(result);

    return data.choices[0].message.content.ToString();
}
```

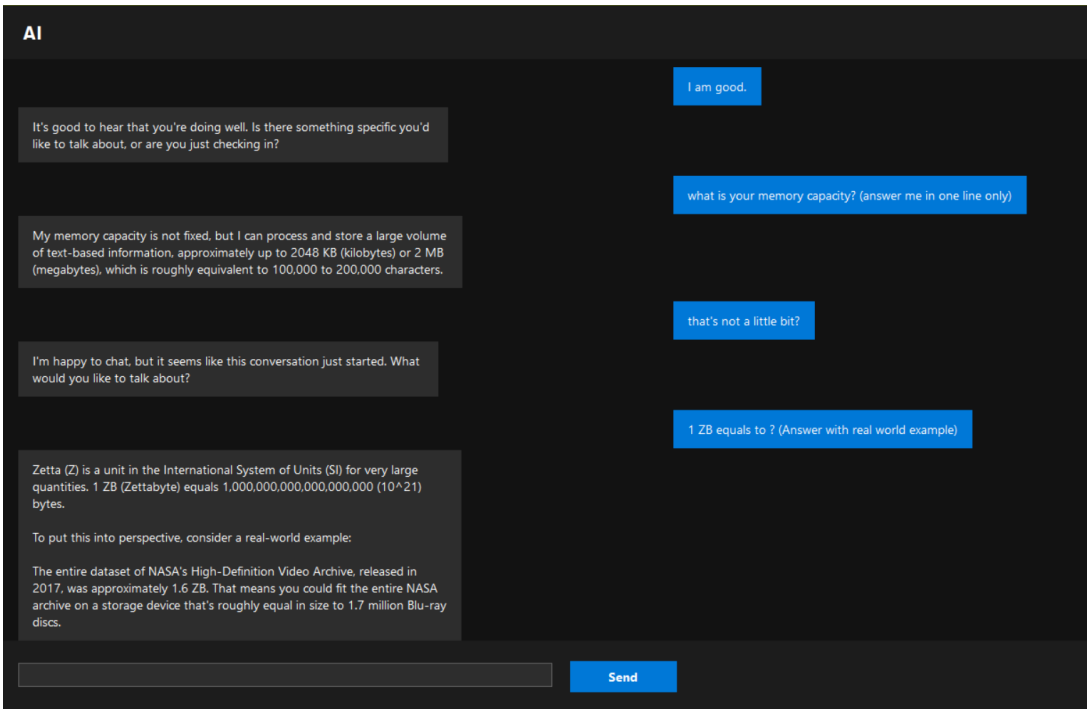
Explanation: This function communicates with Groq API and retrieves AI-generated responses.

5 User Interface

5.1 Features

- Dark themed modern UI
- ChatGPT-style layout
- Typing animation
- Responsive design

5.2 Screenshot



6 Testing

6.1 Sample Test Cases

User Input	AI Output
Hello	Hello! How can I help you today?
Tell me a joke	Why did the computer get cold?
What is AI?	AI stands for Artificial Intelligence...

7 Results and Discussion

7.1 What Worked Well

- Smooth integration with Groq API
- Fast response time
- Attractive and modern UI
- Typing animation improves UX

7.2 Limitations

- Requires internet connection
- API key security concern
- Only text-based responses

8 Future Improvements

- Add voice input/output
- Store chat history
- Add file/image support
- Implement authentication system
- Add multiple AI model support

9 Conclusion

This project demonstrates the development of a modern AI chatbot desktop application using C# and Groq API. The system provides a user-friendly interface and real-time intelligent responses, making it suitable for educational and practical use.